

Is Hubble's Law Merely An Observer Effect?

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Abstract

In this paper, we stated an uncertain principle, explained Hubble's Law from it, and derived dark energy abundance as geometric constant.

Introduction

In 1929, Hubble showed galaxies tend to move away with empirical formula $\langle v \rangle = H_0 r$. However, current explanation is based on Big Bang which implies time has beginning, and dark energy still hasn't been observed. These push us for finding an alternative which is time-symmetric and doesn't rely new parameters.

Uncertain Principle

Although we can just find a metric, we try going a little deeper. We know every object propagates information around, but information also carries momentum! Hence the presence of observer implies

$$\frac{1}{2}T \geq \rho c^2$$

where ρ is known as "vacuum energy density", but actually uncertainty of spacetime.

Hubble's Law

As every toy model, we seek when it reaches minimum everywhere in Newton's approximation. Counting degree of freedom, the energy density is

$$T = \frac{3}{2}\rho c^2$$

Hence the gravitational potential is $V = -\frac{GM}{r}$ where

$$M = \int 4\pi r^2 \frac{3}{2}\rho dr$$

convert to velocity with $v_0 = 0$, we found

$$v = \sqrt{4\pi G\rho r}$$

indeed proportional to r .

Dark Energy

The relation between H_0 and ρ is

$$H_0^2 = 4\pi G\rho$$

compare to $H_0^2 = \frac{8\pi G\rho}{3\Omega_\Lambda}$ from Big Bang, we found

$$\Omega_\Lambda \approx 0.67$$

Think we solely consider “vacuum” solution in classical settings, it’s acceptable.

Anomaly

Despite being successful, the error still couldn’t be omitted. Recall that Dirac also derived gyromagnetic ratio as geometric constant, and QED corrected for anomaly. This motivates us think if our error is also an “anomaly”, and there is a gravity theory beyond GR could correct it.

Reference

- [1] Edwin Hubble, “A Relation between Distance and Radial Velocity among Extra-Galactic Nebulae”
- [2] Franz Mandl & Graham Shaw, “Quantum Field Theory”